

Computer Networks — Introduction & Protocol Stack

GATE-Oriented Notes

1. Computer Network

A **computer network** is a collection of interconnected devices that communicate and share data/resources using defined **protocols**.

The basic objective of networking is to understand:

How does data travel from one computer/process to another?

Examples: Web browsing, email, online banking, file transfer, video streaming, etc.

2. Protocol ★ ★ ★

A **protocol** is a set of rules that governs communication between network entities.

It defines things such as:

- What information is exchanged
- How it is formatted
- When it is sent
- How it is interpreted

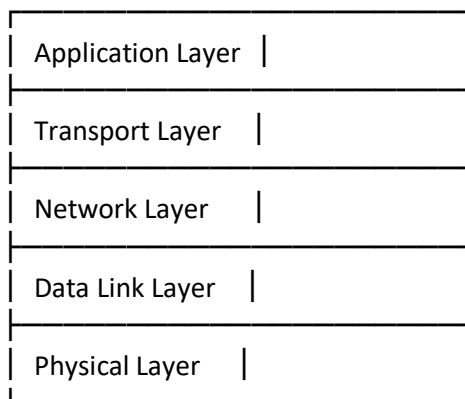
Example: HTTP is used for communication between web clients and servers.

Remember: Protocol = Rules of Communication

3. Layered Network Architecture ★ ★ ★ ★ ★

Networking is divided into multiple **layers**, where each layer performs a specific set of functions.

A commonly used TCP/IP 5-layer representation is:



Layering reduces complexity and allows different protocols/technologies to work together.

4. Physical Layer — Layer 1

Main function:

Transmission of **raw bits** over a physical medium using signals.

Physical media may be:

- Copper cable
- Optical fiber
- Wireless/radio

Devices:

Repeater and Hub operate at/primarily at the Physical Layer.

Repeater

Regenerates/reshapes a degraded signal so that transmission can continue over a longer distance.

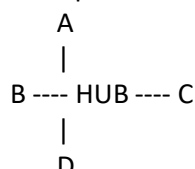
Hub

A multi-port Layer 1 device that forwards incoming signals to other ports.

5. Collision Domain ★ ★ ★

A **collision domain** is a portion of a network in which simultaneous transmissions can interfere/collide.

Example with a hub:



All connected devices share the same collision domain in traditional shared Ethernet.

Collision → Retransmission → Bandwidth wastage

Therefore, reducing collisions improves network efficiency.

6. Data Link Layer — Layer 2

The Data Link Layer provides communication over a **single link/local network** and deals with **frames**.

Important device:

Switch

A Layer-2 switch:

- Separates collision domains.
- Uses link-layer information (e.g., MAC addresses) to forward frames.
- A normal Layer-2 switch does **not** separate the broadcast domain.

Example:



Each switch port generally represents a separate collision domain.

Important distinction ★ ★ ★ ★ ★

Device	Collision Domain	Broadcast Domain
Hub	Same	Same
Layer-2 Switch	Divided	Same
Router	Divided	Divided

Memory trick:

Switch breaks collisions; Router breaks broadcasts.

7. Broadcast Domain ★ ★ ★

A **broadcast domain** is the set of devices that can receive a Layer-2 broadcast.

A Layer-2 switch normally forwards broadcasts, so all its connected ports belong to the same broadcast domain (unless VLANs are used).

A **router separates broadcast domains**.

8. Network Layer — Layer 3 ★ ★ ★ ★ ★

When communication is required between **different networks**, Layer 3 is needed.

Main functions:

- Logical addressing
- Routing
- Packet forwarding between networks

Main device:

Router

Example:

Network A

|

Router

|

Network B

A router can connect multiple networks/interfaces.

Routing

Suppose several paths exist:

Source

|

/\

A B

\ /

C

|

Destination

The network layer determines an appropriate path toward the destination.

Routing vs Forwarding

Routing: Determines/selects the path.

Forwarding: Actually sends the packet through the appropriate outgoing interface/next hop.

This distinction is important for GATE.

9. Internet = Interconnected Networks ★ ★ ★

The Internet is not one single physical network.

It consists of many interconnected networks:

Network A — Router — Network B — Router — Network C

Different networks can be independently managed but communicate using common networking protocols.

Autonomous System (AS)

An **Autonomous System** is a collection of networks/prefixes under a common administrative control and routing policy.

The Internet consists of many such independently managed routing domains.

10. Transport Layer — Layer 4 ★ ★ ★ ★ ★

The Network Layer handles communication between hosts/networks, but a single computer can run many processes/applications simultaneously.

Example:

Computer

|— Browser

|— Email

|— Video application

|— Other processes

So we need communication between **processes**.

Transport Layer:

Provides **process-to-process communication**.

Important transport protocols:

- **TCP**
- **UDP**

Transport layer can also provide mechanisms for:

- Error control
- Flow control
- Congestion control
- Reliable delivery (e.g., TCP)

11. End-to-End Communication

At the source:

Application



Transport



Network



Data Link



Physical

Data travels through the network.

At the destination, the corresponding layers process the data upward:

Physical



Data Link



Network



Transport



Application

Thus, the communication is logically between corresponding layers at the two end systems.

12. Intermediate Devices ★ ★ ★ ★

An important concept:

Intermediate devices do not necessarily process all layers.

Example:

PC A → Switch → Router → Switch → PC B

End systems

Usually use the complete stack:

Application

Transport

Network

Data Link

Physical

Switch

Primarily processes:

Data Link

Physical

Router

Primarily processes:

Network
Data Link
Physical

The higher-layer data is treated as payload from the perspective of a lower/intermediate layer.

13. Encapsulation ★ ★ ★ ★ ★

When data moves down the protocol stack, each layer adds its own control information, typically a header.

Application Data



Transport Header + Data



Network Header + Transport Data



Data Link Header + Network Data



Bits/signals

At the receiver, the reverse process occurs.

Important terms

Layer	Data Unit
Application	Message/Data
Transport	Segment (TCP) / Datagram (UDP)
Network	Packet/Datagram
Data Link	Frame
Physical	Bits

14. Important Protocols ★ ★ ★ ★ ★

Layer	Examples
Application	HTTP, FTP, SMTP
Transport	TCP, UDP
Network	IP, IPv4, IPv6
Data Link	Ethernet, Wi-Fi
Physical	Physical transmission standards/media

Application protocols

HTTP → Web communication

FTP → File transfer

SMTP → Email transfer

Transport

TCP → Reliable, connection-oriented transport

UDP → Connectionless, lightweight transport

Network

IP → Logical addressing and routing/packet delivery across networks.

15. Cross-Layer Protocols

Some protocols don't fit perfectly into a single layer in every textbook/reference.

Examples:

- ARP

- DHCP
- DNS
- SNMP

They may interact with functions associated with multiple layers.

For GATE, focus more on **what the protocol does** rather than blindly memorizing its layer classification.

16. Top-Down vs Bottom-Up Approach ★ ★ ★

Top-Down

Application



Transport



Network



Data Link



Physical

Starts from applications and goes downward.

Bottom-Up

Physical



Data Link



Network



Transport



Application

Starts from physical layer and goes upward.

The lecture follows the:

Top-Down approach



GATE Quick Revision

Remember this table ★ ★ ★ ★ ★

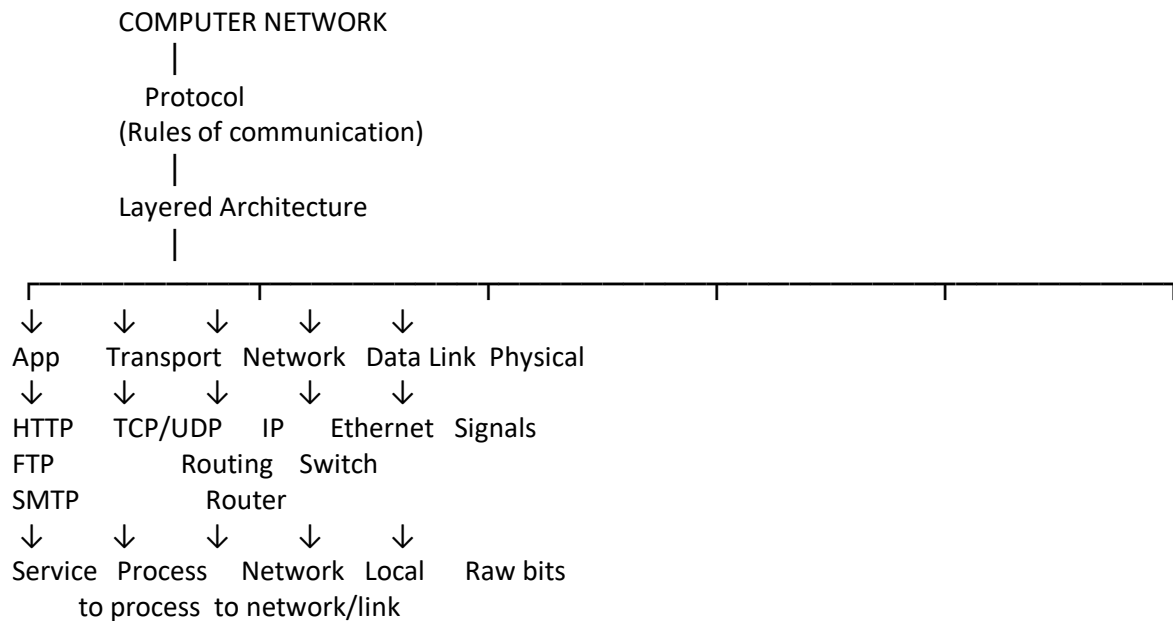
Layer	Core Function	Important Device/Protocol
Application	Network applications/services	HTTP, FTP, SMTP
Transport	Process → Process	TCP, UDP
Network	Routing, logical addressing	IP, Router
Data Link	Link/local delivery, frames	Ethernet, Switch
Physical	Bit/signal transmission	Hub, Repeater

And these 6 facts:

1. Protocol = set of communication rules.
2. Hub → Layer 1.
3. Switch → Layer 2 → separates collision domains.
4. Router → Layer 3 → connects different networks and separates broadcast domains.
5. Transport Layer → **process-to-process** communication.
6. Internet → **interconnected networks** communicating through common protocols.



GATE Concept Map



One-line memory trick

Application = What service? → Transport = Which process? → Network = Which network/path? → Data Link = Which local link/device? → Physical = How to transmit bits?

Computer Networks — Protocol Stack, OSI & TCP/IP

GATE-Oriented Notes

1. Protocol

A **protocol** is a set of rules that controls communication between two or more systems.

A protocol specifies:

- **Sequence** of messages
- **Format/layout** of messages
- Fields and their meaning
- Size/structure of messages
- Rules for sending and receiving messages

Example

HTTP, FTP, SMTP, DNS, DHCP, TCP and IP are protocols used for different networking tasks.

Key Point:

Protocols allow **interoperability** between heterogeneous systems, i.e., different systems/devices can communicate using common standards.

2. Why Layering?

Network communication is complex, so it is divided into **layers**.

Each layer:

- Performs a specific function.
- Provides services to the layer above it.
- Uses services of the layer below it.

General flow

Application



Transport

↓
Network
↓
Data Link
↓
Physical

The data generated by an upper layer becomes the **payload** of the lower layer.

3. OSI Reference Model

OSI = **O**pen **S**ystems **I**nterconnection

It has **7 layers**:

- 7. Application
- 6. Presentation
- 5. Session
- 4. Transport
- 3. Network
- 2. Data Link
- 1. Physical

Functions of OSI Layers

Layer	Main Function	Important Concepts
7. Application	Network services to applications	HTTP, FTP, SMTP, DNS
6. Presentation	Data representation/formatting	Encryption, compression, translation
5. Session	Establish/maintain/terminate sessions	Session management
4. Transport	End-to-end communication	TCP, UDP
3. Network	Packet delivery and routing	IP, routing
2. Data Link	Node-to-node/link communication	Frames, MAC, error detection
1. Physical	Transmission of bits	Signals, physical medium

Layer-wise PDU names

Transport → Segment / Datagram
 Network → Packet
 Data Link → Frame
 Physical → Bits

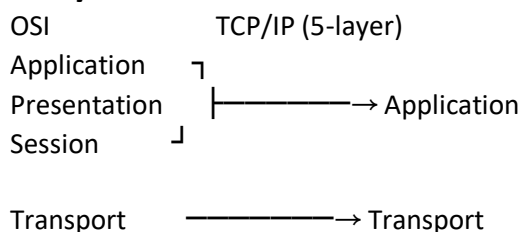
4. TCP/IP Protocol Stack

TCP/IP is the dominant protocol suite used for Internet communication.

A commonly used **5-layer representation** is:

Application
 Transport
 Network
 Data Link
 Physical

TCP/IP vs OSI



Network _____ → Network

Data Link _____ → Data Link

Physical _____ → Physical

Thus, in the common 5-layer TCP/IP model:

OSI Presentation + Session functions are included in the Application layer.

Important Note

Some textbooks represent TCP/IP as a **4-layer model** by combining:

Data Link + Physical

into a single **Network Access/Link layer**.

So don't get confused if you see **4-layer or 5-layer TCP/IP** representations.

5. Important Protocols by Layer

Application Layer

- HTTP/HTTPS → Web
- FTP → File transfer
- SMTP → Email transfer
- DNS → Domain Name System
- DHCP → Dynamic host configuration

Transport Layer

- TCP
- UDP
- RTP

Network Layer

- IPv4
- IPv6

Data Link Layer

- Ethernet
- Wi-Fi
- Bluetooth

Cross-layer / Supporting Protocols

- ARP
- DHCP
- DNS
- SNMP

6. Encapsulation ★ ★ ★ ★ ★

When data moves **down the protocol stack**, each layer adds its own control information, usually a header.

Example:

Application:

DATA

Transport:

[T Header | DATA]

Network:

[N Header | T Header | DATA]

Data Link:

[DL Header | N Header | T Header | DATA]

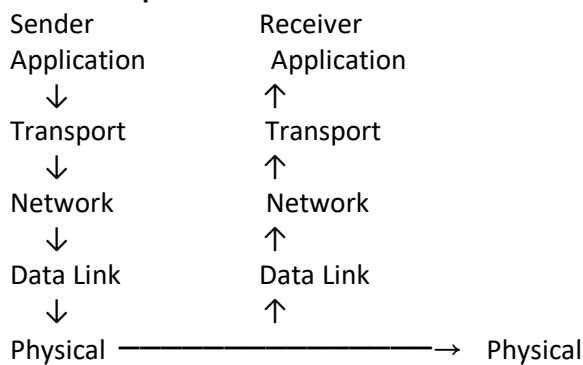
Physical:

Bits

This process is called **Encapsulation**.

At the receiver, the reverse process occurs:

Decapsulation



GATE Point

Sender: headers are added while moving downward.

Receiver: corresponding headers are processed/removed while moving upward.

7. Network Communication Levels

Different layers provide communication at different levels.

Network Layer

Provides **host-to-host** communication.

Host A —————→ Host B

Transport Layer

Provides **process-to-process** communication.

Process A

↓

Network

↓

Process B

Example: A computer may simultaneously run a browser, email application and another network application. Transport-layer mechanisms help deliver data to the appropriate process using transport-layer addressing such as port numbers.

★ Remember

Network Layer → Host-to-Host

Transport Layer → Process-to-Process

8. Reliability at Different Layers

Network Layer

IP provides **best-effort/unreliable delivery**.

It does not guarantee:

- Successful delivery
- Correct ordering
- Retransmission

Transport Layer

Can provide:

- **Reliable delivery → TCP**
- **Unreliable/connectionless delivery → UDP**

Therefore:

Network → generally unreliable/best effort

Transport → may provide reliable or unreliable service

9. Network Devices and Their Layers



Different networking devices operate primarily at different layers.

Device	Layer	Main Address/Information
Repeater	Layer 1	Physical signals
Hub	Layer 1	Physical signals
NIC	Mainly Layer 2	MAC address
Bridge	Layer 2	MAC address
Switch	Layer 2	MAC address
Router	Layer 3	IP address

Easy memory:

Hub → Signal

Switch → MAC

Router → IP

10. Repeater

A repeater is a **Layer 1 device**.

Its basic purpose is to regenerate/strengthen signals so that communication can cover a greater physical distance.

It does not understand MAC or IP addresses.

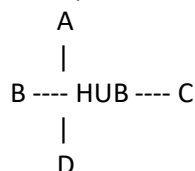
11. Hub

A hub is essentially a **multi-port repeater**.

Characteristics

- Layer 1 device
- Does not understand MAC addresses
- Cannot intelligently filter traffic
- Connected devices share the medium
- Causes a shared collision domain

Example:



If A transmits, the signal is repeated to other ports.

Key Point

Hub = Layer 1 + shared medium + one collision domain

12. Bridge

A bridge operates at **Layer 2**.

It can examine **MAC addresses** and filter/forward frames accordingly.

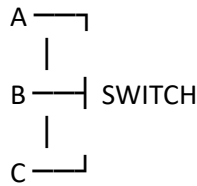
Therefore traffic can be localized instead of unnecessarily being sent everywhere.

13. Switch ★ ★ ★ ★ ★

A switch is essentially a **multi-port Layer 2 bridge**.

It uses **MAC addresses** to forward frames.

Example:



If the switch knows the destination MAC address, it can forward the frame through the appropriate port.

Key Points

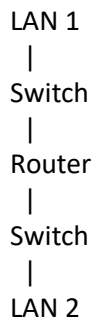
- Layer 2 device
- Uses MAC addresses
- Forwards frames
- Each switch port represents a separate collision domain
- A normal switch does **not** break a broadcast domain by itself

14. Router ★ ★ ★ ★ ★

A router primarily operates at **Layer 3 (Network layer)**.

It uses **IP addresses** and routing information to forward packets between different networks.

Example:



Main functions

- Connects different networks
- Routes/forwards packets
- Uses IP addresses
- Separates broadcast domains

Key Point

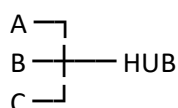
Router → Layer 3 → IP → Inter-networking

15. Collision Domain ★ ★ ★ ★ ★

A **collision domain** is a network segment in which simultaneous transmissions can interfere/collide.

Hub

All devices connected to a hub share the same collision domain.



→ One shared collision domain

Switch

Each switch port forms a separate collision domain.

A — Port 1

B — Port 2

C — Port 3

SWITCH

Thus:

Switch separates collision domains.

16. Broadcast Domain ★ ★ ★ ★ ★

A **broadcast domain** is the set of devices that receive a Layer-2 broadcast.

Switch

A normal Layer-2 switch forwards broadcasts within the same broadcast domain.

Therefore:

Switch does not normally break a broadcast domain.

Router

A router does not normally forward Layer-2 broadcasts from one interface to another.

Therefore:

Router breaks/separates broadcast domains.

17. Collision Domain vs Broadcast Domain

Device	Collision Domain	Broadcast Domain
Hub	One shared domain	One
Switch	Each port is a separate collision domain	Usually one
Router	Separates network segments	Separates broadcast domains

★ GATE Shortcut

Hub → Collision domain NOT divided

Switch → Collision domain divided

Router → Broadcast domain divided

18. LAN and WAN

LAN — Local Area Network

Network covering a relatively small geographical area.

Examples:

- Home
- Office
- College lab
- Building

WAN — Wide Area Network

Network covering a large geographical area and connecting geographically separated sites.

Examples:

Office A ——— WAN ——— Office B

WAN connectivity is often provided through service providers.

Remember:

LAN → Local/small area

WAN → Wide/large area

19. Important Historical Evolution

Exact dates are **low priority for GATE**, but the broad evolution is useful:

Telegraph



Telephone networks



Packet Switching



ARPANET



TCP/IP



DNS



WWW



Modern Internet

Important historical concepts

- **ARPANET** → precursor to today's Internet
- **Packet switching** → data divided into packets
- **TCP/IP** → foundation of Internet inter-networking
- **DNS** → translates domain names to IP-related information
- **WWW** → World Wide Web

Don't waste time memorizing every date mentioned in the lecture.

20. Packet Switching ★ ★ ★

In packet switching, large data is divided into smaller **packets**.

Large Data



P1 | P2 | P3 | P4

Packets may travel through different routes and are processed at the destination.

Key idea

Packet switching allows network resources to be shared efficiently among multiple communications.

Circuit switching and packet switching will be studied separately in detail.



GATE Most Important Points

Must remember:

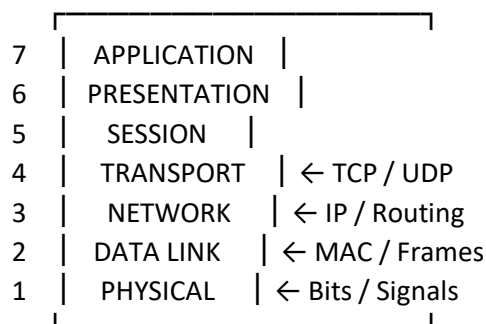
1. **OSI** → 7 layers
2. **TCP/IP** → commonly 5 layers
3. TCP/IP may also be represented as a **4-layer model**
4. OSI Presentation + Session → generally included in TCP/IP Application layer
5. **Protocol** = rules + message sequence + message format
6. **Encapsulation** → downward
7. **Decapsulation** → upward
8. **Physical** → bits/signals

9. Data Link → frames + MAC
10. Network → packets + IP + routing
11. Transport → end-to-end/process-to-process
12. TCP → reliable
13. UDP → connectionless/unreliable
14. Hub → Layer 1
15. Switch → Layer 2 → MAC
16. Router → Layer 3 → IP
17. Hub → shared collision domain
18. Switch → separates collision domains
19. Router → separates broadcast domains
20. Network layer → host-to-host
21. Transport layer → process-to-process
22. IP → best-effort/unreliable
23. LAN → local area
24. WAN → large geographical area

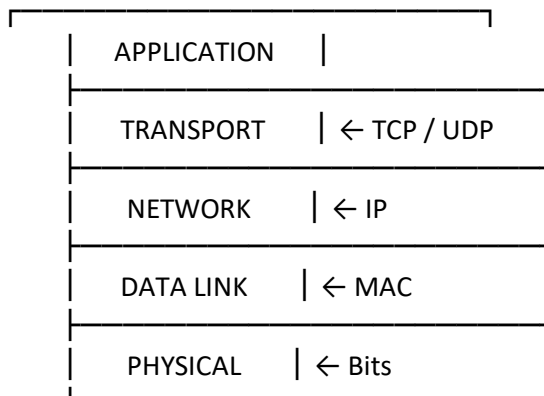


One-Minute Revision Diagram

OSI MODEL



TCP/IP MODEL



DEVICE QUICK REVISION

Hub → L1 → Signal
 Switch → L2 → MAC
 Router → L3 → IP

COMMUNICATION

Network Layer → Host-to-Host
 Transport Layer → Process-to-Process

GATE Focus

Highest priority:

OSI/TCP-IP layers → layer functions → encapsulation → TCP/UDP → IP → MAC →
Hub/Switch/Router → collision domain → broadcast domain → host-to-host vs process-to-process.

Lower priority:

Internet history and exact historical dates.

~Anshu